Lecture 9: introduction to atomics

read-modify-write, get-and-add, compare-and-swap, spin lock, lock-free stack, ABA problem

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https://github.com/Svazars/parallel-programming/blob/main/slides/pdf/l9.pdf

In previous episodes

Formalization of concurrent execution

- Timeline, Event, Interval, Precedence

Concurrent objects

- Sequential object, Sequential specification, Linearizability, Linearization points

Progress conditions

- Dependent progress: Deadlock-freedom, Starvation-freedom
- Non-blocking progress: Lock-freedom, Wait-freedom
- Dependent non-blocking progress: Obstruction-freedom

Register design space

- Bool/Int, SRSW/MRSW/MRMW, Safe/Regular/Atomic
- Atomic snapshot

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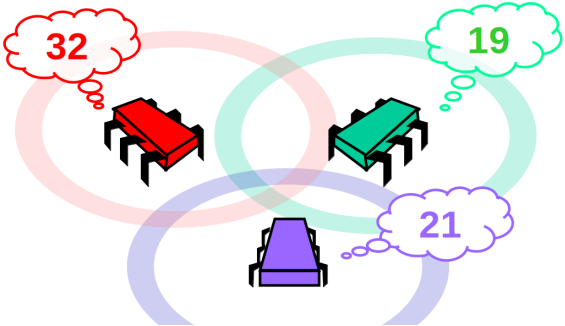
Let's switch to practical aspects.

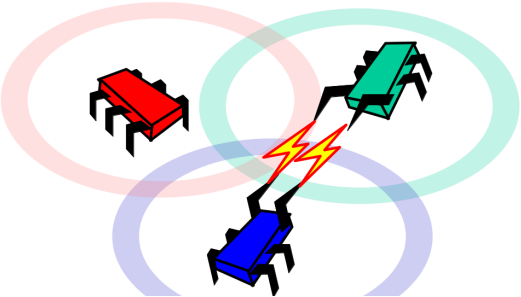
Lecture plan

1. Consensus
2. Read-Modify-Write
3. Concurrent Counter: puzzlers
4. Basic spin locks
5. Lock-free stack and ABA
6. Summary

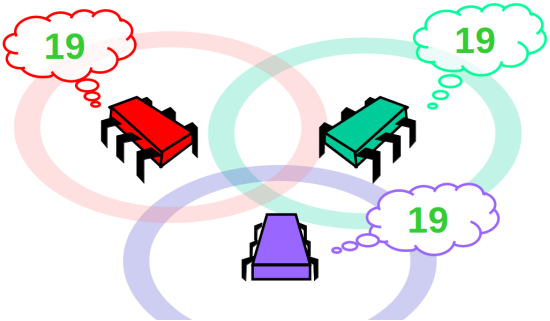
Each thread has private input



They communicate



They agree on some input



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- Asynchronous computability different from Turing computability

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Read-write registers formalized in terms of safe/regular/atomic concurrent objects.

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Read-write registers formalized in terms of safe/regular/atomic concurrent objects.
Theorem could be adapted to:

- Registers
- Message-passing
- Carrier pigeons
- Any kind of asynchronous computation

Consensus number

An object **X** has *consensus number* **n**

- If it can be used to solve **n**-thread consensus
  - Take any number of instances of **X**
  - together with atomic read/write registers
  - and implement **n**-thread consensus
- But not **(n+1)**-thread consensus

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Atomic read/write registers have consensus number 1

Atomic registers cannot implement multiple assignment

- Single write/multi read OK
- Multi write/multi read impossible

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Consensus numbers measure synchronization power

Theorem

- *If you can implement* $\boldsymbol{X}$ *from* $\boldsymbol{Y}$
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Registers have consensus number **1**

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Registers have consensus number **1**

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Registers have consensus number **1**
There are practically interesting problems that require higher consensus number
What should we do next?

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Read-Modify-Write Objects

Method call

- Returns previous value x
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public abstract class RMWRegister {
  private int value;
  public int synchronized getAndMumble() {
    int prior  = value;
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RMW everywhere!

- Most synchronization instructions are RMW methods
- The rest can be trivially transformed into RMW methods

Read-Modify-Write: read

```
public abstract class RMWRegister {
  private int value;
  public int synchronized read() {
    int prior  = value;
    value = value; // mumble == identity
    return prior;
  }
}
```

Read-Modify-Write: getAndSet

```
public abstract class RMWRegister {
  private int value;
  public int synchronized getAndSet(int v) {
    int prior  = value;
    value = v; // mumble(x) = v, constant
    return prior;
  }
}
```

Read-Modify-Write: getAndIncrement

```
public abstract class RMWRegister {
  private int value;
  public int synchronized getAndIncrement() {
    int prior  = value;
    value = value + 1; // mumble(x) = x + 1
    return prior;
  }
}
```

Read-Modify-Write: getAndAdd

```
public abstract class RMWRegister {
  private int value;
  public int synchronized getAndAdd(int a) {
    int prior  = value;
    value = value + a; // mumble(x) = x + a
    return prior;
  }
}
```

Read-Modify-Write: compareAndSet

```
public abstract class RMWRegister {
  private int value;
  public boolean synchronized compareAndSet(int expected, int update) {
    int prior = value;          // load witness value
    if (value == expected) { // if current value equals to expected
      value = update;           // then replace it with update
      return true;              // and return with success
    }
    return false;               // else return with failure, change nothing
  }
}
```

Read-Modify-Write: compareAndExchange

```
public abstract class RMWRegister {
  private int value;
  public int synchronized compareAndExchange(int expected, int update) {
    int prior = value;          // load witness value
    if (value == expected) { // if current value equals to expected
      value = update;           // then replace it with update
    }
    return prior;               // return witness value
  }
}
```

Read-Modify-Write: now you know it

AtomicInteger[1]:

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AtomicInteger[1]:

- `get()`, `getAndSet(int)`

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Atomic*[2]

- AtomicBoolean
- AtomicInteger, AtomicIntegerArray
- AtomicLong, AtomicLongArray
- AtomicReference, AtomicReferenceArray

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Atomic RMW: what's the point?

General form of atomic RMW:

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- No wait-free implementation of RMW registers from atomic registers
- Hardware RMW instructions not just a convenience

Foundations: here we go again

Atomic registers

- From safe SRSW Boolean register

Foundations: here we go again

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- From safe SRSW Boolean register up to atomic MRMW Integer register

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When will we stop?

Consensus: final point

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`getAndSet`, `getAndIncrement`, `getAndAdd` *have consensus number exactly 2*

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Theorem

`getAndSet`, `getAndIncrement`, `getAndAdd` *have consensus number exactly 2*

Theorem

`compareAndSet` *has* ∞ *consensus number*

Consensus: final point

Theorem

`compareAndSet` has ∞ consensus number

Consensus: final point

Theorem

compareAndSet has ∞ consensus number

```
public class RMWConsensus {
  private AtomicInteger r = new AtomicInteger(-1); // init with -1
  private int[] proposed = new int[N]; // values proposed by threads
  public int decide(int value) {
    int i = ThreadID.get();    // I am thread i
    proposed[i] = value;       // I propose value
    r.compareAndSet(-1, i);    // try to set my id as `winner`
    return proposed[r.get()]; // return value proposed by winner
  }
}
```

The consensus hierarchy

- **1**. Read/Write Registers, Snapshots ...
- **2**. `getAndSet`, `getAndIncrement` ...
- ...
- ∞. `compareAndSet` ...

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Some other environments – distributed systems, supercomputers, telegraph, pigeon posts etc. – could be **different**. Now you know how to do the analysis of "computability" in such environments.

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Challenges of the new era:

- Highly distributed systems (internet nodes, geographically distributed data centers) with non-guaranteed message delivery (UDP)
- Heterogeneous computing (CPU, GPU, FPGA, GPGPU ...)

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SynchronizedCounter

```
public class SynchronizedCounter {
    private long cnt;
    public SynchronizedCounter(long initial) { this.cnt = initial; }
    // assume delta > 0
    public synchronized void increment(int delta) { cnt += delta; }
    public synchronized long get() { return cnt; }
}
```

- Wait-free, Lock-free, Obstruction-free?
- Starvation-free, Deadlock-free?

FairLockedCounter

```
public class FairLockedCounter {
    private long cnt;
    private final Lock lock = new ReentrantLock(/* fair = */ true);
    public FairLockedCounter(long initial) { this.cnt = initial; }
    public void increment(int delta) {
      lock.lock() try { cnt += delta; } finally { lock.unlock(); }
    }
    public long get() {
      lock.lock(); try { return cnt; } finally { lock.unlock(); }
    }
}
```

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AtomicLoopCounter

```
public class AtomicLoopCounter {
    private long cnt;
    private final AtomicBoolean status = new AtomicBoolean(false);
    public AtomicLoopCounter(long initial) { this.cnt = initial; }
    public void increment(int delta) {
      while (status.getAndSet(true) == true) {}
      try { cnt += delta; } finally { status.set(false); }
    }
    public long get() {
      while (status.getAndSet(true) == true) {}
      try { return cnt; } finally { status.set(false); }
    }}
```

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AtomicIncCounter

```
public class AtomicIncCounter {
    private final AtomicLong cnt = new AtomicLong(0);
    public AtomicIncCounter(long initial) { cnt.set(initial); }
    public void increment(int delta) {
      cnt.getAndAdd(delta);
    }
    public long get() {
      return cnt.get();
    }}
```

- Wait-free, Lock-free, Obstruction-free?
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AtomicCASCounter

```
public class AtomicCASCounter {
    private final AtomicLong cnt = new AtomicLong(0);
    public AtomicCASCounter(long initial) { cnt.set(initial); }
    public void increment(int delta) {
      long expected = cnt.get();
      while (true) {
        long witness = cnt.compareAndExchange(expected, expected + delta);
        if (witness == expected) { return; } else { expected = witness; }
      }
    }
    public long get() { return cnt.get(); }}
```

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ReplicatedCounter

```
static final ThreadLocal<Long> privateCounter = new ThreadLocal<Long>();
public void increment(int delta) {
  privateCounter.set(privateCounter.get() + delta);
}
public long get() {
  long result = 0;
  for (Thread t: ALL_THREADS) result += t.privateCounter.get(); // PSEUDOCODE
  return result;
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- Linearizable? What if `ThreadLocal` are Atomic MRMW registers?

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- Linearizable? What if `ThreadLocal` are Atomic MRMW registers?

Reminder: Task 2.4, NegativeSplitCounter

Read-modify-write operations not always are linearization points

https://github.com/Svazars/parallel-programming/tree/main/hw/block2/9.1

Homework

Task 9.1

- *`HWQueue.enq` consists of two instructions*
- *Prove that first instruction (`getAndIncrement`) is not a linearization point in some valid executions*
- *Prove that second instruction (`set`) is not a linearization point in some valid executions*
- *Consider provided counterexamples: do they prove that `HWQueue` is not linearizable?*

Concurrent counters: summary

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Lecture plan

1. Consensus
2. Read-Modify-Write
3. Concurrent Counter: puzzlers
4. Basic spin locks
5. Lock-free stack and ABA
6. Summary

Test-And-Set: non-reentrant boolean spin lock

```
class TASlock {
  private static final boolean LOCKED = true, UNLOCKED = false;
  private final AtomicBoolean state = new AtomicBoolean(UNLOCKED);
  void lock() {
    while (true) {
      boolean before = state.getAndSet(LOCKED); // <-----------|
      if (before == UNLOCKED) { return; } // I win             |
      else {}                             // I lose, repeat ---+
  }}
  void unlock() { state.set(UNLOCKED); }
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```

- Pros: easy to implement, easy to prove correctness, ultra-fast ownership handoff
- Cons: burn CPU, contention on memory bus

Test-And-Set: reentrant spin lock

https://github.com/Svazars/parallel-programming/tree/main/hw/block2/9.2

Homework

Task 9.2

- *Replace* `AtomicBoolean` *with* `AtomicStampedReference(Thread.currentThread, counter)` *and make TAS lock reentrant.*
- *Provide at least 3 tests written in JCStress.*

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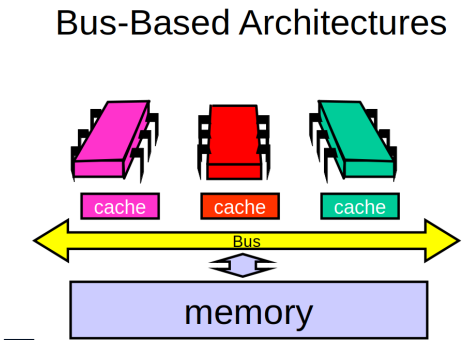
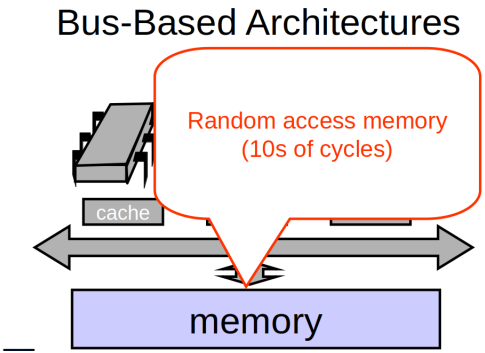
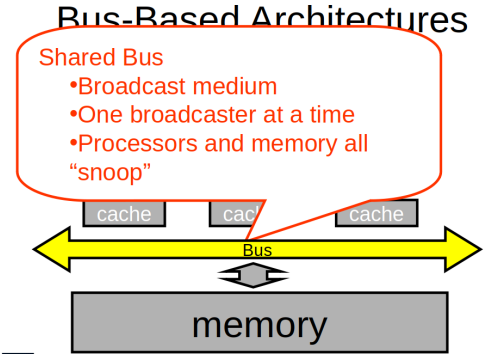
Memory bus



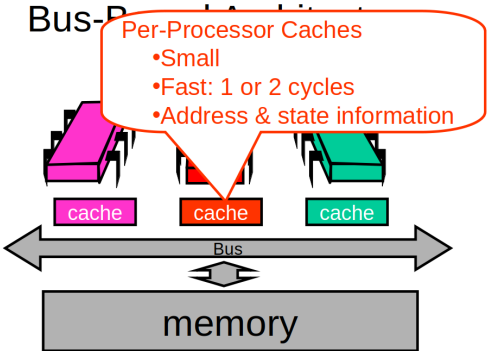
Memory bus


Bus-Based Architectures
Random access memory
(10s of cycles)
cache
memory

Memory bus


Bus-Based Architectures
Shared Bus
•Broadcast medium
•One broadcaster at a time
•Processors and memory all "snoop"
cache
cache
cache
Bus
memory

Memory bus


Per-Processor Caches
•Small
•Fast: 1 or 2 cycles
•Address & state information
cache
cache
cache
Bus
memory

Cache

- **Cache hit** – data found in local cache of current processor
- **Cache miss** – up-to-date data must be found elsewhere

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There are many questions related to cache-based design:

- How many level of caches should be used?
- What is optimal size for each level?
- How to properly synchronize caches?
- How to move out stale data from cache?
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We will go deeper in next Lecture.

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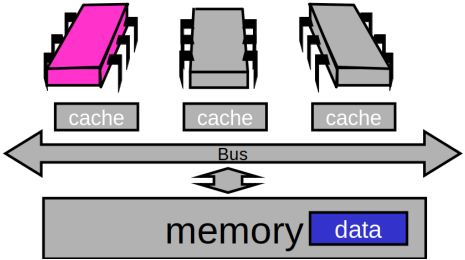
Today we will try to illustrate that even basic low-level knowledge helps to speed-up concurrent algorithms.

Read data

Processor Issues Load Request



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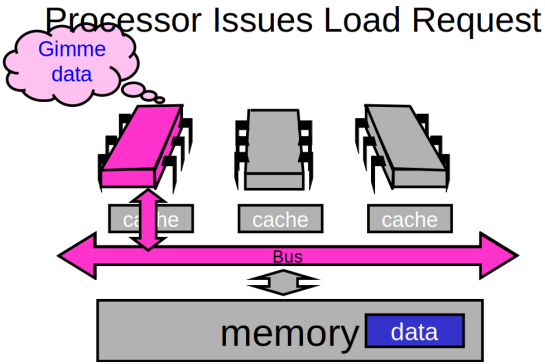
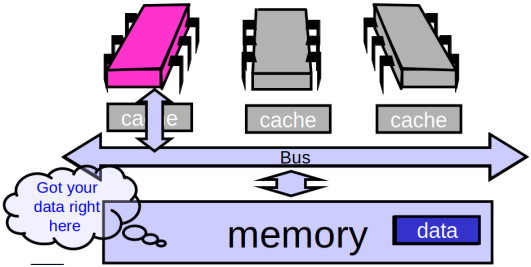
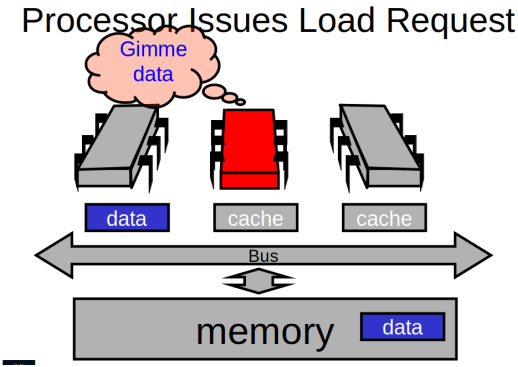

Processor Issues Load Request
Gimme data
cache
cache
cache
Bus
memory
data

Read data

Memory Responds



Read data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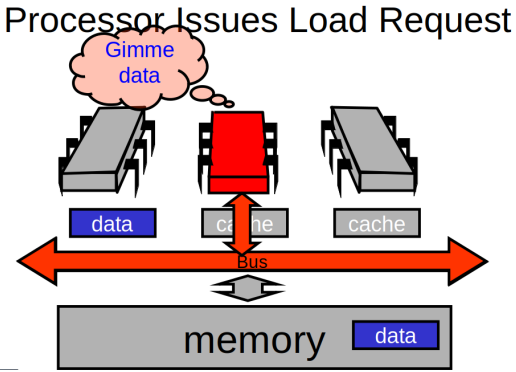
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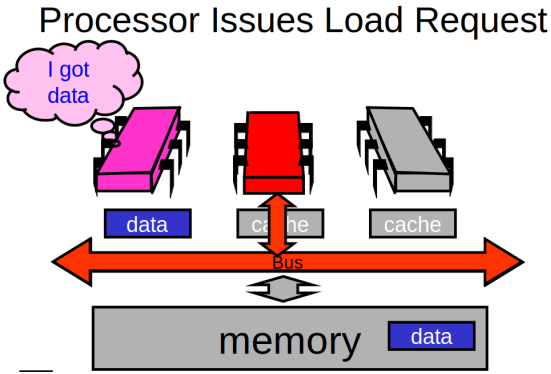
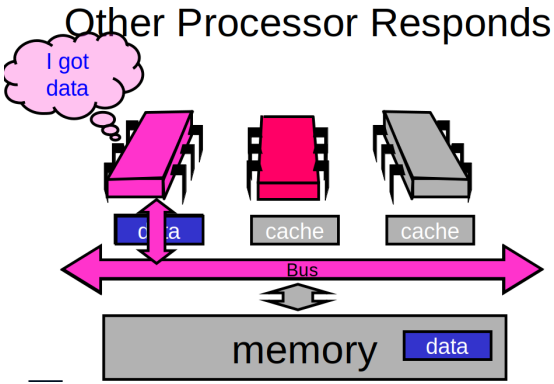
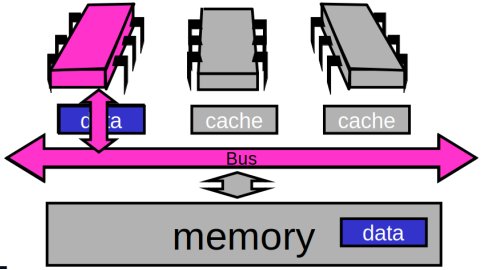

Processor Issues Load Request
I got data
data
cache
cache
Bus
memory
data

Read data


Other Processor Responds
I got data
cache
cache
Bus
memory
data

Read data

Other Processor Responds



Read data

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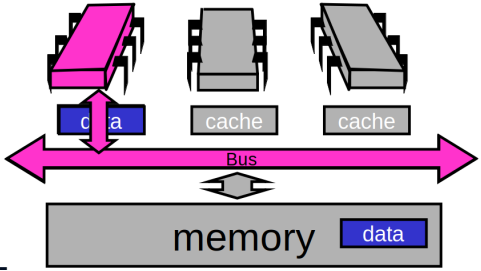
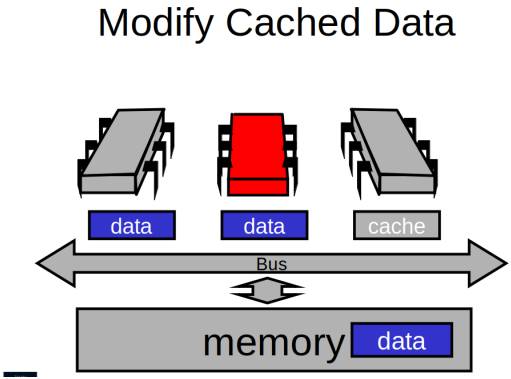


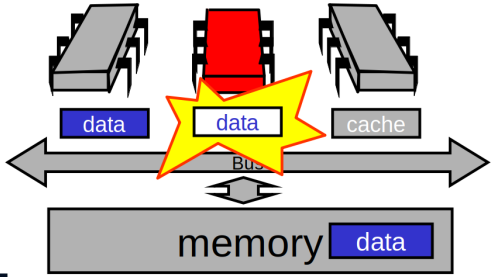
And now we have several copies of the same data in different places

Modify data



Modify data

Modify Cached Data



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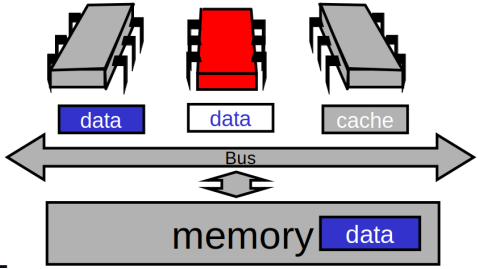

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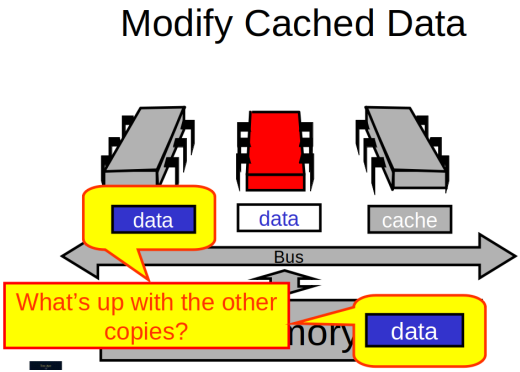
Modify Cached Data
data
data
cache
Bus
What's up with the other copies?
data

Cache coherence

We have lots of copies of data

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- Cached copies at processors

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Let's go back to the naive multiprocessor model.

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Cache coherence: message passing

Messages

- Request for read, Respond with data
- Request for invalidate, Acknowledge invalidation
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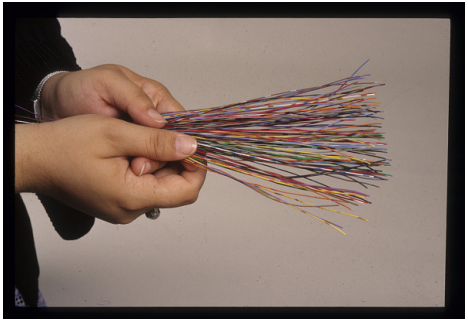
Nanoseconds[3]:



[3] https://americanhistory.si.edu/collections/object/nmah_692464

Memory bus, cache coherence, multiprocessor: takeaways

Multiprocessor systems have complicated memory subsystem

Memory bus, cache coherence, multiprocessor: takeaways

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Simplified cost model:

- Read from private non-shared data is ultra-fast
- Write to private non-shared data is ultra-fast
- Read from private shared data is ultra-fast
- Write to shared data requires communication (invalidate and await)
- Read/write of non-cached location takes significant time

TASLock: reminder

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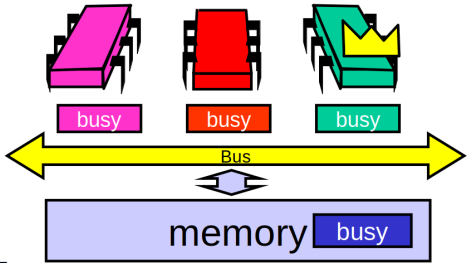
What happens on release?
Invalidation storm.

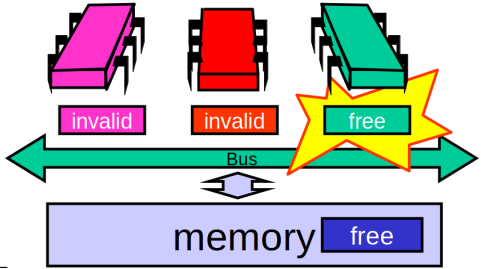
Invalidation storm

Local Spinning while Lock is Busy



Invalidation storm

On Release



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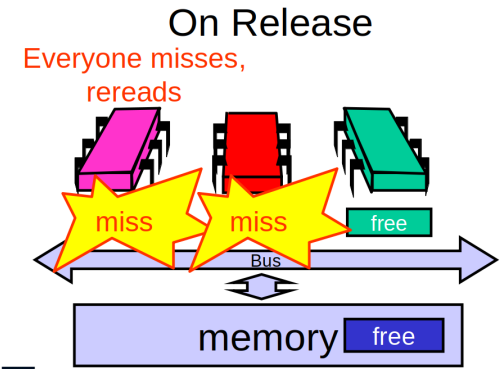
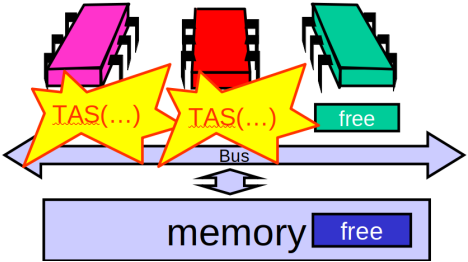


Invalidation storm

On Release

Everyone tries TAS



TATAS: invalidation storm

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General pattern for TATAS:

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General pattern for TATAS:

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- Pause without using bus

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- If pause > quiescence time
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General pattern for TATAS:

- Acquire lock
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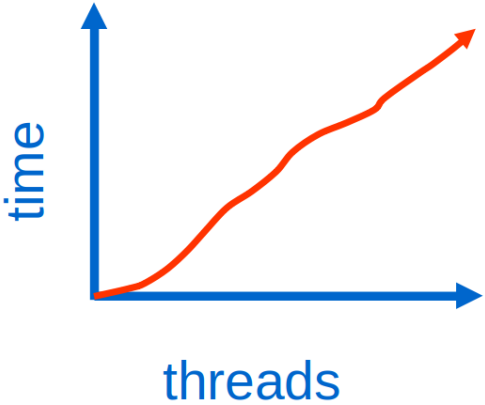
- Use bus heavily
- If pause > quiescence time
  - critical section duration independent of number of threads
- If pause < quiescence time
  - critical section duration slower with more threads

Quiescence Time



Increases linearly with the number of processors for bus architecture

Spin locks and hardware, you know so far

- Easy to implement and prove correctness
- Burn CPU for better responsiveness
- Read before trying to acquire lock, writes to actually shared memory are expensive
- Suffer from invalidation storm in contended environment when lock is released

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- Maybe application is misdesigned?

Any kind of "storm" (trap storm, invalidation storm, scheduler storm etc.) is a strong sign of misdesign. You could workaround it to some extent, but it will manifest itself later (more threads, faster h/w, better OS).
Anyway, let's try to avoid the invalidation storm and (maybe) improve our skills with "designing it right"

Exponential Backoff

```
public void lock() {
  int delay = MIN_DELAY;
  while (true) {
    while (state.get() == LOCKED) {} // read-only polling
    if (lock.getAndSet(LOCKED) == UNLOCKED) return; // return on success
    sleep(random() % delay); // delay next write request on failure
    delay = Math.min(delay * 2, MAX_DELAY); // keep delay within bounds
  }
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```

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- Trade-off: CPU overuse vs. latency
- Still very naive, we will continue in Lecture 12 (queue locks)

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- Burn CPU vs. better responsiveness
- Actual performance depend on h/w (cache coherency) and application (contention)

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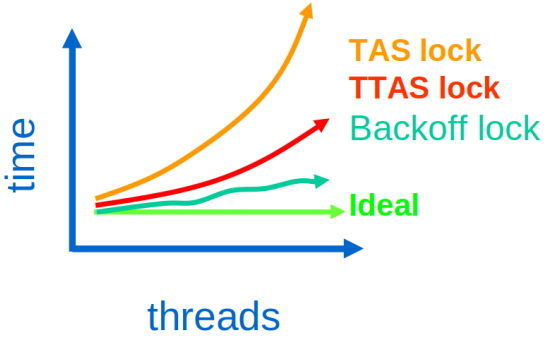

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- Spin locking is a design pattern, building block for "normal" locking algorithms

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- Spin locking is a design pattern, building block for "normal" locking algorithms
- **NEVER** use plain spin locks anywhere in production system[4]

[4] https://www.realworldtech.com/forum/?threadid=189711&curpostid=189723

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Critically important:

- Spin locking is a design pattern, building block for "normal" locking algorithms
- **NEVER** use plain spin locks anywhere in production system[4] unless you know what you are doing. After finishing this introductory course on concurrency, you definitely **don't**.

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Spin locks: homework

`https://github.com/Svazars/parallel-programming/tree/main/hw/block2/9.3`

Homework

Task 9.3 Measure performance of `TASLock`, `TATASLock`, `ExpBackoffLock`

- *1, 2, 3, 4, 5, 6, 7, 8, 16, 32 concurrent threads*
- *with small critical section (increment)*
- *with moderate critical section (compute n-th fibonacci number recursively)*

using JMH.

Lecture plan

1. Consensus
2. Read-Modify-Write
3. Concurrent Counter: puzzlers
4. Basic spin locks
5. Lock-free stack and ABA
6. Summary

Lock-free stack

- `push(x)`
- `pop()`
- Last-in, First-out (LIFO) order

Lock-free stack: push

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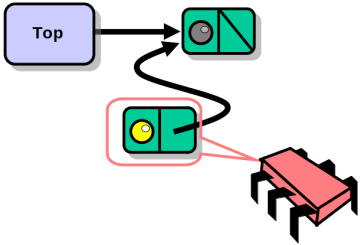

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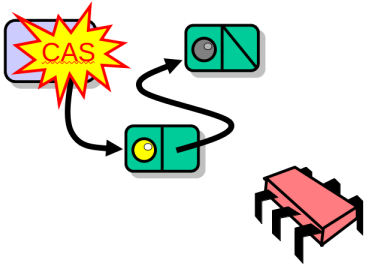

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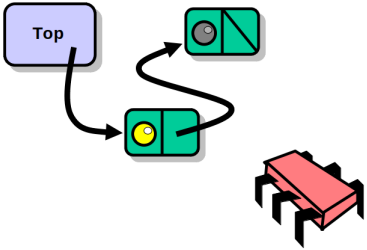

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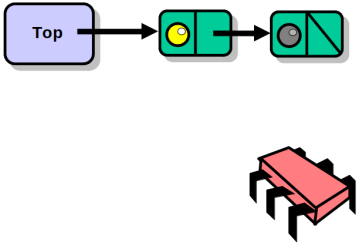

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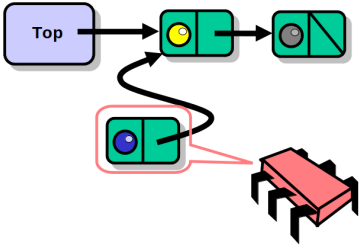
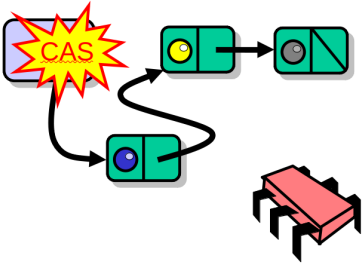


Lock-free stack: push



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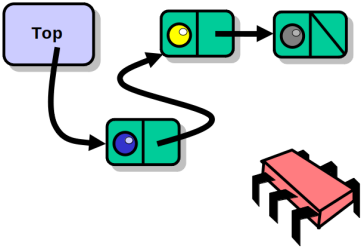
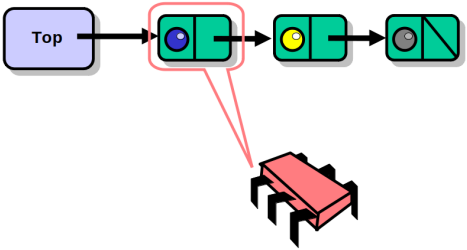


Lock-free stack: pop



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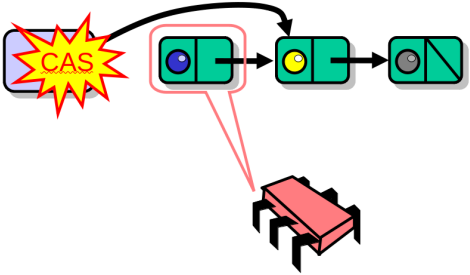

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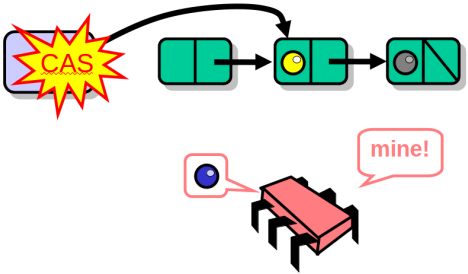

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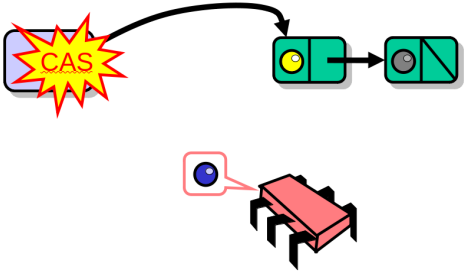

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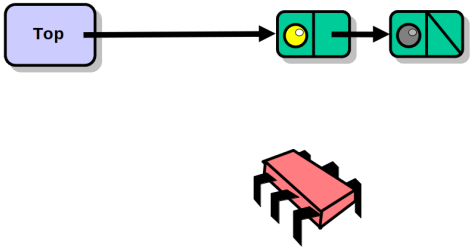

Lock-free stack

```
public class LockFreeStack {
  private AtomicReference top = new AtomicReference(null);
  public boolean tryPush(Node node){
    Node oldTop = top.get();
    node.next = oldTop;
    return top.compareAndSet(oldTop, node);
  }
  public void push(T value) {
    Node node = new Node(value);
    while (true) {
      if (tryPush(node)) return;
      backoff.backoff();
  }}
```

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  }}
```

- Wait-free, lock-free or obstruction-free?

Lock-free stack: summary

- Your first non-trivial and useful lock-free algorithm based on RMW
- Elegant
- Easy to reason about
- Easy to define linearization points

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But not perfect?

Node pooling

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  }
  public void push(T value) {
    Node node = new Node(value); // Yikes!



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  }
  public void push(T value) {
    Node node = ThreadLocalNodePool.get(); // successful `pop`
                                           // will populate this cache
    node.value = value;
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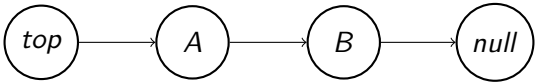
```

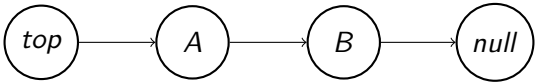
ABA problem



Thread 1

Thread 2

ABA problem



Thread 1 Thread 2

Start `pop()` (`top.CAS(A, B)`)

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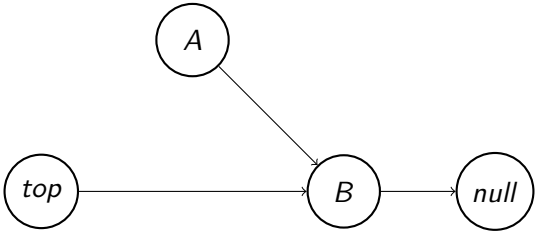

Thread 1

Start `pop()` (`top.CAS(A, B)`)

Thread 2

`A = pop()`

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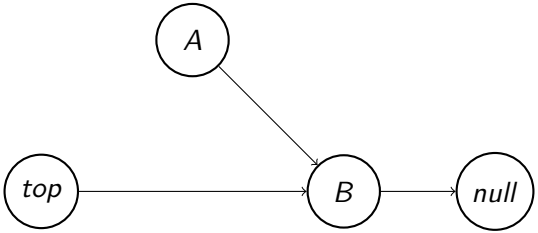

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```
A = pop()
reclaim(A)
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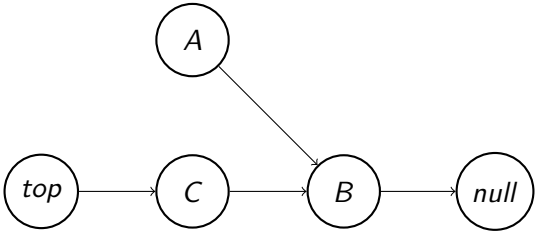


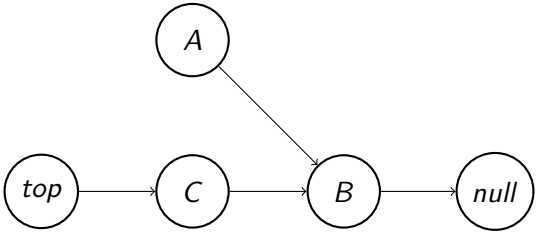
Thread 1

Start `pop()` (`top.CAS(A, B)`)

Thread 2

```
A = pop()
reclaim(A)
push(C)
```

ABA problem



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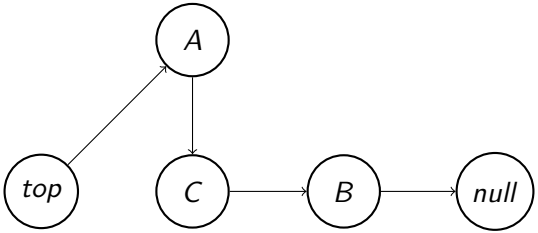

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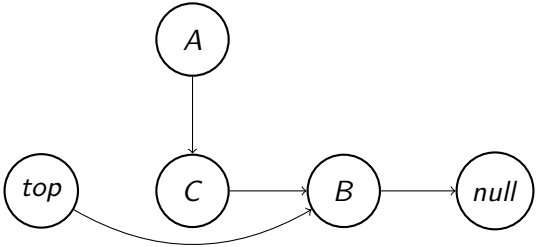

Thread 1
Start `pop()` (`top.CAS(A, B)`)
Finish `pop()`

Thread 2

```
A = pop()
reclaim(A)
push(C)
push(A')
```

ABA problem

ABA problem is always a headache in non-GC languages.
It still happens in managed languages, be aware.
Advanced algorithms you could or could not use:

- `AtomicStampedReference`[5]
- immortal memory
- RCU
- reference counting
- other variation of GC
- hazard pointers
- ...

[5] https://docs.oracle.com/en/java/javase/11/docs/api/java.base/java/util/concurrent/atomic/AtomicStampedReference.html

Summary

Read-Modify-Write atomic operations

- `getAndSet`, `getAndAdd` – consensus number **2**
- `compareAndSet`, `compareAndExchange` – consensus number ∞

Spin locks

- Test-and-set
- Test-and-test-and-set
- Exponential Backoff

Memory hierarchy

- Multi-level caching
- Cache coherence via replication and message passing
- Invalidation storm

Lock-free stack and ABA for pooling.

Summary: homework

`Lecture 9, Tasks 9.*:` https://github.com/Svazars/parallel-programming/blob/main/hw/block2